# Data Source Evaluation Report

How ready your sources were, and what was left unmapped

**Date:** 2026-11-11

**Project:** pc-insurance-claims-fraud

## Summary

- Total data sources: 3
- Mapped: 3
- Needs mapping: 0
- Needs enrichment: 1

All sources remain below the 70% quality threshold after safe format normalization, and all three are mapped. The user chose to proceed with SIU_WATCHLIST as-is after review found no further safe normalization; its 56.5% score and completeness limitations are accepted for now. No duplicate (DATA_SOURCE, RECORD_ID) pairs or PERSON-name type candidates were found. One SIU_WATCHLIST record has no name attribute; this row-level gap is noted below. Original files are preserved; the registry points to derived -improved.jsonl mapping inputs. Ambiguous slash dates remain verbatim in the improved files.

The quality visualization and screenshot are the original Step 6 baseline, before normalization; the updated scores and gate below are authoritative for the improved files.

## Data Source Details

### Data Source 1: POLICYHOLDERS

**Status:** Mapped

**Location:** data/raw/POLICYHOLDERS-improved.jsonl

**Original:** data/raw/POLICYHOLDERS.jsonl

**Records:** 120

**Fields:** 26 distinct FEATURES attributes; 6 root fields, including DATA_SOURCE and RECORD_ID

**Evaluation:**

- Contains 114 PERSON and 6 ORGANIZATION records. The shared-field type-applicability check passed.
- Quality changed from 51.2% to 51.5% after consistent casing was applied to mixed coded values: completeness 33.2%, format consistency 92.9%, and duplicate-key rate 0.0%.
- FEATURES attributes match the current Entity Specification. Root fields AMOUNT, CATEGORY, DATE, and STATUS are preserved as payload in the mapped output.
- The source uses DATA_SOURCE=CUSTOMERS; the mapped output uses the registered project code POLICYHOLDERS.
- Explicit or unambiguous DATE and DATE_OF_BIRTH values, phone punctuation, one complete SSN

format, outer whitespace, and mixed casing in coded values were normalized. Ambiguous slash dates, identifier lengths, and phone lengths/country-code notation were preserved.

**Reason:** Mapping is complete, but the score remains below 70% because of source-data completeness gaps.

**Record type check:** 0 of 114 PERSON-typed records are type/name candidates. Decision: none needed.

**At the time of this mapping assessment:** SIU_WATCHLIST mapping remained to be completed; no records had yet been loaded. Reviewer approval was later confirmed before Module 6 loads.

### Data Source 2: VENDOR_REGISTRY

**Status:** Mapped

**Location:** data/raw/VENDOR_REGISTRY-improved.jsonl

**Original:** data/raw/VENDOR_REGISTRY.jsonl

**Records:** 22

**Fields:** 30 distinct FEATURES attributes; 5 root fields, including DATA_SOURCE and RECORD_ID

**Evaluation:**

- Contains 16 PERSON and 6 ORGANIZATION records. The shared-field type-applicability check passed.
- The initial formatting pass changed quality from 56.3% to 56.4% after address-country casing was standardized (38.5% completeness, 97.7% format consistency, 0.0% duplicate rate). A later mapping-review iteration normalized one display-name/address wrapper in EMAIL_ADDRESS to its enclosed source-supported address; the original collected file remains unchanged. The updated score is 57.0% (38.5% completeness, 100.0% format consistency, 0.0% duplicate rate).
- FEATURES attributes match the current Entity Specification. Root CATEGORY, DATE, and STATUS remain payload.
- The input uses DATA_SOURCE=REFERENCE; the mapped output uses the registered project code VENDOR_REGISTRY.
- Recognizable date-of-birth layouts, phone punctuation, address-country casing, outer whitespace, and one email wrapper were normalized. Postal codes, country aliases, tax IDs, phone lengths, and LEIs were left unchanged because their differing values may be substantive.

**Reason:** Mapping is complete, but the score remains below 70% because of source-data completeness gaps.

**Record type check:** 0 of 16 PERSON-typed records are type/name candidates. Decision: none needed.

**At the time of this mapping assessment:** All three sources were mapped and the VENDOR_REGISTRY output was ready. The later Module 6 load is documented in docs/loading_strategy.md.

### Data Source 3: SIU_WATCHLIST

**Status:** Mapped; proceeding with documented quality limitations

**Location:** data/senzing-ready/SIU_WATCHLIST.jsonl

**Original:** data/raw/SIU_WATCHLIST.jsonl

**Mapping input:** data/raw/SIU_WATCHLIST-improved.jsonl

**Records:** 17

**Fields:** 19 distinct FEATURES attributes; 5 root fields, including DATA_SOURCE and RECORD_ID

**Evaluation:**

- Contains 17 PERSON records. The shared-field type-applicability check passed.
- Quality changed from 56.2% to 56.5% after category casing was standardized: completeness 39.5%, format consistency 95.3%, and duplicate-key rate is 0.0%.
- FEATURES attributes match the current Entity Specification. Root fields CATEGORY, DATE, and STATUS are preserved as payload.
- The source label WATCHLIST is aligned with the registered project code SIU_WATCHLIST in the mapped output.
- One of 17 records has no name attribute. Explicit or unambiguous dates, phone punctuation, category casing, and outer whitespace were normalized; ambiguous slash dates, differing year widths, and phone lengths were preserved.
- The validated 17-record mapping output was ready at the location above before Module 6; the later load is documented in docs/loading_strategy.md.

**Reason:** Mapping is complete, but quality remains below 70% because of sparse source data. The single record without a name is a data gap, not a type/name candidate. No further safe normalization was identified; do not impute values.

**Record type check:** 0 of 17 PERSON-typed records are type/name candidates. Decision: none needed.

**Next step at assessment time:** Continue to Data processing with the current mapped output. The 56.5% score and missing-name gap were documented; the source was later loaded as-is at the bootcamper's direction.

## Mapping Priority

Completed mapping order; this is not approval to load records.

1. POLICYHOLDERS - establishes the insurer's core policyholder population and is the largest source. 2. VENDOR_REGISTRY - connects claim-paid businesses with associated people and organizational relationships. 3. SIU_WATCHLIST - adds the screening identities used to surface potential SIU leads; preserve and explicitly handle the one record with no name.

## Cross-Source Outlook

Counts below are exact distinct-value overlaps on the named attributes, not counts of matching records or resolved entities. They were remeasured from the improved copies after normalization.

- POLICYHOLDERS × VENDOR_REGISTRY - measured
  - PHONE_NUMBER: 1 distinct value shared (of 14 in POLICYHOLDERS, 7 in VENDOR_REGISTRY).

    - NAME_FIRST: 0 shared (of 79, 2); NAME_LAST: 0 (of 59, 2); DATE_OF_BIRTH: 0 (of 37, 2); EMAIL_ADDRESS: 0 (of 13, 2).

- POLICYHOLDERS × SIU_WATCHLIST - measured
    - NAME_LAST: 6 distinct values shared (of 59 in POLICYHOLDERS, 9 in SIU_WATCHLIST); NAME_FIRST: 5 (of 79, 8); DATE_OF_BIRTH: 3 (of 37, 5); SSN_NUMBER: 2 (of 5, 2); EMAIL_ADDRESS: 1 (of 13, 4).
    - PHONE_NUMBER: 1 shared (of 14, 2).
- VENDOR_REGISTRY × SIU_WATCHLIST - measured
    - PHONE_NUMBER: 0 distinct values shared (of 7 in VENDOR_REGISTRY, 2 in SIU_WATCHLIST); NAME_FIRST: 0 (of 2, 8); NAME_LAST: 0 (of 2, 9); EMAIL_ADDRESS: 0 (of 2, 4); DATE_OF_BIRTH: 0 (of 2, 5).

## Quality Gate

Re-scoring uses quality_score = 0.70 × completeness + 0.25 × format_consistency + 0.05 × (100 - duplicate_rate). The mapping-stage results are 51.5% for POLICYHOLDERS, 57.0% for VENDOR_REGISTRY, and 56.5% for SIU_WATCHLIST. Each remains below 70%. Completeness gaps were not filled; no values were invented. The improved files retain all 159 source records and original keys. All three sources were subsequently loaded; validation is documented in docs/results_validation.md.

The weakest coverage fields recorded in the existing assessment remain completeness gaps: POLICYHOLDERS (NAME_TYPE 1/120, NAME_MIDDLE 2/114, NATIONAL_ID_COUNTRY 3/120); VENDOR_REGISTRY (NATIONAL_ID_TYPE, NATIONAL_ID_NUMBER, and NATIONAL_ID_COUNTRY, each 2/22); SIU_WATCHLIST (NAME_MIDDLE, GENDER, and DRIVERS_LICENSE_NUMBER, each 1/17, tied with other fields). These missing values cannot be safely filled in this module.

The initial Step 7b pass found no further safe mechanical normalization at that point. A later VENDOR_REGISTRY review identified one source-supported email-format correction. The current sample-data catalog confirms these collected sources match the fixed Senzing Demo Truth Set (159 records across CUSTOMERS, REFERENCE, and WATCHLIST). For SIU_WATCHLIST, no additional safe correction remained; the user chose to proceed with the current mapped data despite its quality limitations. Remaining completeness gaps were not filled by inference.

## Safe Normalization Applied

- Trimmed outer whitespace without altering internal characters.
- Standardized only explicit or unambiguous dates to MM/DD/YY or MM/DD/YYYY, preserving whether the original year had two or four digits. Two-digit years were not expanded to a guessed century; ambiguous slash dates were restored verbatim, and unambiguous day-first values were converted without changing the date.
- Removed phone-number punctuation while preserving digits and an existing leading +; did not infer missing digits or country codes.
- Standardized casing in mixed coded-value fields while preserving spelling and aliases.
- Normalized one email display-name/address wrapper to the enclosed address, which is independently source-supported; the original collected file remains unchanged.

- Formatted complete nine-digit SSNs with standard separators. Did not pad or truncate identifiers, unify country aliases, or fill missing values.
- Kept original fetched files unchanged and retained their fetch validation metadata in the registry; those validation fields refer only to each original_file_path.
- Remeasured format consistency with the same field-dominant-pattern method used for the initial scores. All record counts and (DATA_SOURCE, RECORD_ID) keys are unchanged.
- Audited the final copies: all changed values are formatting-only; ambiguous slash dates remain exactly as collected.